

A SAFE AND X-RAY TRANSPARENT ALUMINIUM SAMPLE CELL FOR HIGH PRESSURE AND HIGH TEMPERATURE NANO-DIFFRACTION IMAGING

A-L. Buisson*, P. Brumund, Y. Watier, H. Djazouli, ESRF, Grenoble, France
M-I. Richard, K. Olson, CEA, Grenoble, France

Abstract

The European Synchrotron Radiation Facility beamline ID01 performs Bragg Coherent Diffraction Imaging and X-ray nano-diffraction experiments with in-situ environments. A new sample environment for nanoparticles has been designed and tested. It combines a furnace and a pressurized X-ray-transparent chamber, without the use of beryllium for safety reasons.

The required 180° horizontal and 45° vertical viewing angles necessitate the use of a dome-shaped pressure chamber. Early in the design phase, the thermal effects of the 500 °C furnace on the dome's mechanical properties were identified as critical. A Finite Element Analysis (FEA) study was conducted, accounting for heat sources, gas turbulence, and static pressure.

Aluminium 6082-T6 was chosen for the pressurized dome, providing a safe and easy-to-procure solution. A dome thickness of 0.5 mm provides 80 % X-ray transmission at 33 keV. The sample, a 200 nm-diameter palladium nanoparticle, reaches the super-critical state at 340 °C in a 45 bar hydrogen atmosphere.

Unlike beryllium-based pressure domes, this design uses aluminium, avoiding machining difficulties, procurement issues, and safety hazards.

INTRODUCTION

ID01 at the European Synchrotron Radiation Facility (ESRF) is a nano-diffraction and coherent imaging beamline [1]. Recent in-situ studies of hydrogen insertion in palladium nanoparticles in subatmospheric pressure and up to 550 °C renewed interest in exploring the high pressure region of the palladium-hydrogen phase diagram. Indeed understanding a reaction at a single catalyst particle level is a prerequisite for understanding the behavior of ensembles of particles because of unavoidable dispersion of non-homogeneous samples. Examples of such sample environment include a compact ultrahigh vacuum chamber [2], gas flow cells [3, 4], and liquid environmental X-ray cells [5–8]. Previous Bragg Coherent Diffraction Imaging (BCDI) studies of the Palladium-Hydrogen system have been limited to low pressure (<1 bar relative) and ambient temperatures [9] missing the opportunity to explore new phenomena at more extreme conditions.

The goal was then to upgrade an existing furnace (operating at 900 °C and below 1 bar relative) [4] to a sample cell capable of withstanding high pressures (up to 50 bar) and

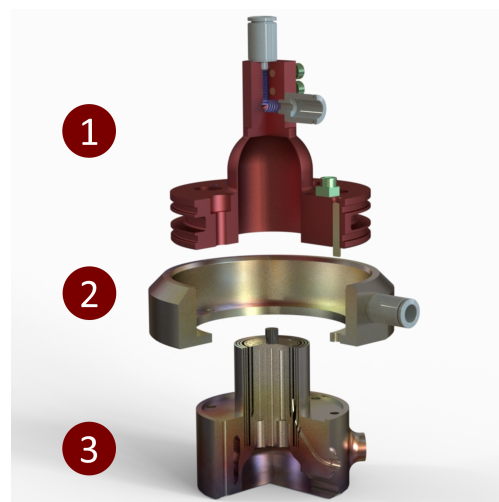


Figure 1: Typical sample cell: (1) Dome chamber, (2) Outer cooling loop, (3) Furnace

high temperatures (up to 500 °C) with an x-ray transmission goal of 80 % at 33 keV. The dome-shaped chamber was chosen for its axial-symmetry required for 180° scanning. As illustrated in Fig. 1 the basic design of a sample cell consists of

1. a removable dome cover that encloses the sample and lets the x-rays pass while permitting rotation around the vertical axis,
2. outer water cooling for the base of the dome a connection to the top of the dome and
3. a heated base (furnace) onto which the sample is mounted with gas inlet and outlet and internal base water cooling loop.

Material choice for dome cell is crucial. While dome cells are often made with beryllium for its high x-rays transmission properties, this material presents significant challenges in terms of procurement and safety due to its toxicity. Ensuring the safety of such pressurized equipment involves validating various steps, including: analytical calculations, Finite Element Analysis (FEA) simulations, water and helium leak tests, outer shape metrology report. Once these steps are validated, the system can be coupled with a furnace and a gas panel, enabling experiments in both static pressure or dynamic flow.

* anne-lise.buisson@esrf.fr

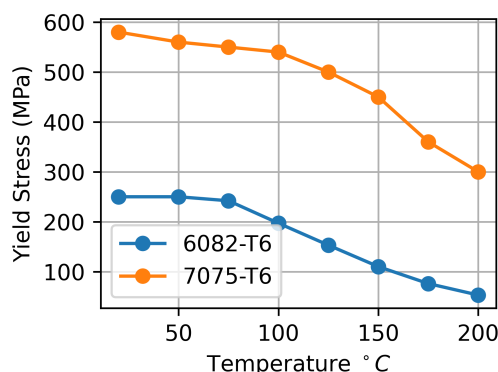


Figure 2: Yield stress variation with temperature for selected aluminum alloys [11, 12].

DESIGN PHASE

Material Choice

The material choice is primarily driven by safety considerations. In terms of optimal x-ray transmission at 20 keV and 33 keV, a beryllium dome would be the best option. However, it significantly increases the risk of toxic exposure. The second best transparent material could be a polymer such as polyether ether ketone (PEEK). While PEEK offers x-ray transparency, its yield stress limit drops drastically with higher temperatures and has poor thermal conductivity, making it incompatible with high temperature experiments. Other common metallic materials, such as copper and stainless steel, lack sufficient x-ray transparency. Titanium has very poor thermal conductivity. Aluminum, however, stands out as the best option, balancing both transparency and thermal conductivity, while eliminating toxic risks. Aluminum such as *Al6082-T6* and *Al7075-T6* are commonly used at the ESRF and are known to be easy to manufacture and procure. *Al6082-T6* is recommended by the French construction code for pressurized equipments: CODAP [10] for its ductility and is therefore our choice.

Dome Thickness

Regarding manufacturing, we decided to avoid 3D printing methods to prevent defects in the structure. Lathe machining is the best option for an aluminium dome shape. Standard machining techniques limit the thickness of the dome to 0.4 mm. The aluminum *Al6082-T6* maximum allowable stress is calculated to be the ultimate stress divided by a safety factor of 2.4 [10], giving the value of 145 MPa. This means that for a cylindrical diameter of 28 mm at 50 bars, the minimum required thickness is 0.48 mm. Finite Element Analysis (FEA) results for a dome with a 28 mm diameter, a 0.5 mm thickness, and a 50 bar pressure load show a maximum stress of 133 MPa (below the limit of 145 MPa), validating the design of thickness 0.5 mm. The total material thickness crossed by the photon beam is therefore 1 mm, resulting in 80 % photon transmission at 33 keV.

Additionally, to address the thermal issues arising from furnace heating, it is clear from Fig. 2 that above 100 °C the

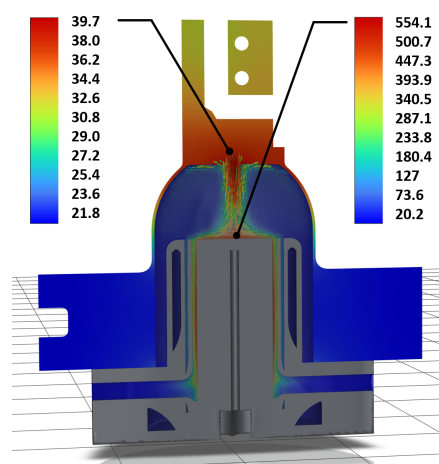


Figure 3: Thermal analysis for the furnace at 150 W and a He pressure of 50 bar.

loss of elastic mechanical properties becomes concerning. Mechanical creep must also be considered, which becomes significant for *Al6082-T6* above 150 °C [13]. Therefore, the water-cooling pipes should be designed to maintain a maximum aluminum dome temperature of 100 °C dictated by the maximum acceptable drop of yield stress.

Regarding the mechanical stress of 133 MPa at 50 bar pressure and maximum temperature of 100 °C the dome will operate with a comfortable safety factor of about $S = \sigma_y(100\text{ °C})/\sigma \approx 1.5$.

Water Cooled Circuit

To prevent exceeding material limits regarding creep and yield strength of *Al6082-T6* in the thinner part of the dome, it is crucial to position the water cooling loop carefully to maintain the temperature below 100 °C. Preliminary calculations of a dome equipped with a simple water-cooling loop at its base indicate that the radiation power transmitted from the furnace to the dome is minimal, accounting for only 3 % of the total power. The primary source of thermal transmission is convection, representing 75 % of the total power, which transfers heat from the sample to the top of the dome, creating a hot-spot at 210 °C. Introducing a cooling zone at the top reduces the aluminum temperature below 100 °C.

The final analysis (Fig. 3), performed with helium gas, shows a maximum temperature of 40 °C, with a furnace power of 150 W to be able to reach 500 °C at the sample position.

Furnace Design

The ID01 furnace requires long-term thermal stability, making heat transfer between the furnace and the sample stage base a key point. Since 2017, ESRF stainless steel furnaces have addressed this by using 3D-printed Direct Metal Laser Sintering (DMLS) bases with integrated water cooling, improving thermal isolation and allowing low-temperature seals. As shown in Fig. 4, Gas and water connections are built into the stainless steel printed body, while heater elements

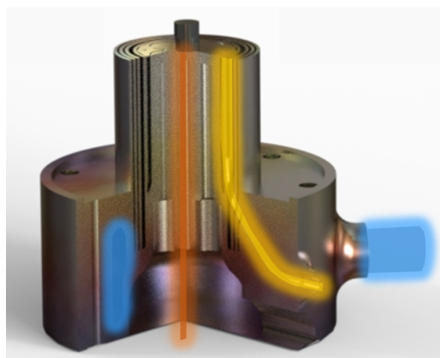


Figure 4: Furnace's section view. Hydrogen flow in yellow, water cooling loop in blue, heaters in orange.

and thermocouples are positioned outside the high-pressure chamber, inside the central pillar, simplifying feedthroughs and replacement of delicate components such as heaters.

One drawback is heat loss from having water-cooled and 500 °C-heated areas in the same part, mitigated by minimizing material thickness. To further reduce convection losses at pressures up to 50 bar, 0.2 mm stainless steel baffles are added around the pillar. The sample is fixed with diluted alumina paste for optimal thermal transfer.

RESULTS

Water and Gas Test

Before any gas usage, it is mandatory to test the furnace and dome cell using water injection at a test pressure of 71.5 bar Bars giving a safety coefficient of 1.43 [10] against ultimate stress. This test ensures that the cell can safely operate at a pressure of 50 bar with a cell temperature of 500 °C. Following this, the cell is tested with helium at 50 bar to detect potential leaks.

Comparative Metrology

To prevent a cell explosion during operation, the cell's dimensions must be measured before and after each experiment. Any deviation should be reported. The preferred technique is caliper measurement or FARO arm, which offers lower measurement error of 10 µm, it enables to measure only on the cylindrical part of the cell, also known to be the weakest area.

Temperature Measurement

For a 150 W furnace, finite element analysis (FEA) predicted a furnace temperature of 554 °C with helium gas at 50 bar. However, operational tests at same power measured a lower furnace temperature of 494 °C. At the sample position, FEA anticipated the sample temperature to match the furnace temperature. In contrast, test results (Fig. 5) show a sample temperature of 340 °C for a setpoint of 390 °C, indicating partial heat conduction loss between the furnace and the sample.

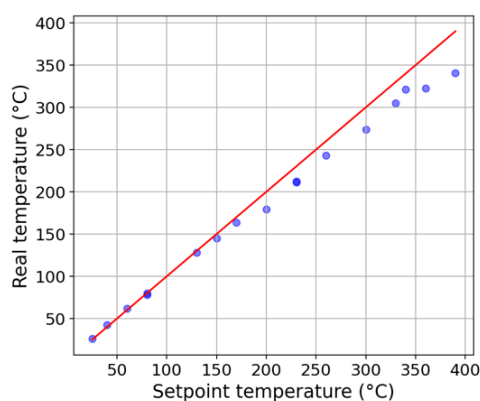


Figure 5: Measured temperatures of a Palladium nanoparticle (blue dots) compared to furnace setpoint (red line).

CONCLUSION

Conducting an experiment involving hydrogen at high pressure and high temperature requires numerous safety precautions. In our case, we first adhered to a construction code for pressurized equipment (CODAP) [10] in compliance with the European Pressure Equipment Directive 2014/68/EU. Aluminium 6082-T6 was chosen over beryllium, providing a safe, easy-to-procure solution.

Analytical calculations were then validated using finite element analysis (FEA), which considered both thermal and static loads of a 500 µm thick dome cell. This FEA guided the design of the water-cooling loop, which is essential to prevent creep and yield stress drops in the aluminum cell by maintaining the dome temperature at 40 °C (target from material limits: below 100 °C).

The dome was manufactured with a nominal thickness of 500 µm which was verified by our metrology lab to be about 490 µm. This is above the accepted minimal value of 480 µm dictated by the pressure vessel standard regarding stress limits. Plastic deformation is monitored between each experiment using both a caliper and a FARO arm, to detect any deformation exceeding 10 µm and mitigate explosion risks. Regarding the experimental results, the furnace went to 50 bar with 100 mbar increments with no outer shape deformation. For temperature results, particle measurements remain accurate up to 175 °C. Beyond this temperature, deviations from the setpoint increase slowly. At the final setpoint of 390 °C, palladium was measured at 340 °C enough to reach the super-critical phase of the Palladium Hydrogen system.

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